

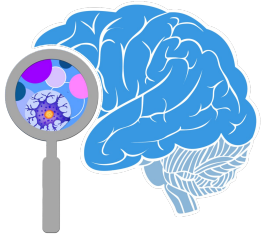
Gas exchange

Lesson objectives:

- To state the gases that make up the air.
- To describe the structure of alveoli.
- To explain how the adaptations of the parts of the gas exchange system help them perform their function.

Which fact is **true**?

- a) There are about **600 million alveoli** in your lungs and if you stretched them out, they would cover an entire **tennis court**.
- b) There are about **1 million alveoli** in your lungs and if you stretched them out, they would cover an entire **football pitch**.
- c) There are about **10 million alveoli** in your lungs and if you stretched them out, they would cover an entire **surface of the Moon**.



The changing composition of air

The table shows the **composition** of inhaled air and exhaled air.

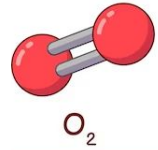
Gas	Inhaled air	Exhaled air
oxygen	21%	16%
carbon dioxide	0.04%	4.0%
water vapour	0.96%	3.0%
nitrogen	78.1%	78.1%
noble gases	0.94%	0.94%

What **similarities** and **differences** can you see?

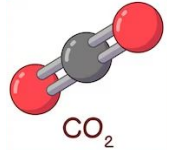
What is gas exchange?

It is the process of gases exchanging at the alveoli. Two gases exchange places.

Oxygen goes from alveoli to blood circulation



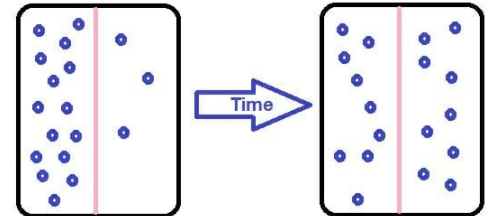
Carbon dioxide goes from blood circulation to alveoli



Aim for excellence!

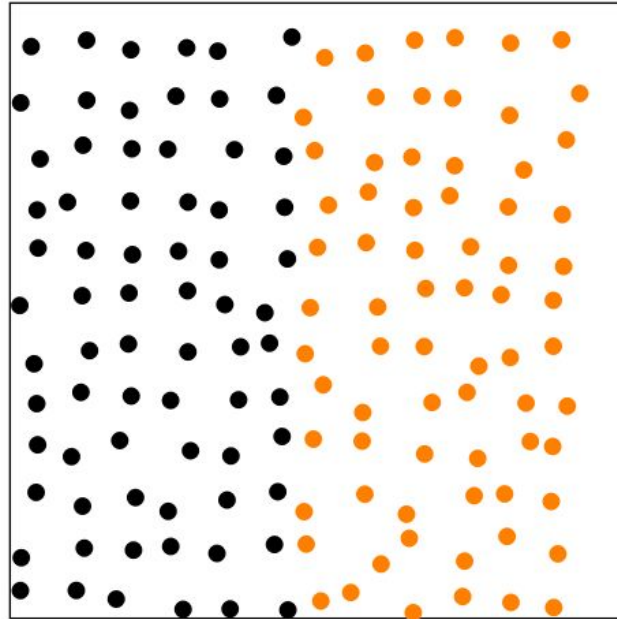
Use the word '**diffuses**' rather than 'goes'

Diffusion is the net movement of particles from a region of high concentration to a region of low concentration.



Diffusion

Diffusion is the movement of particles from an area of high concentration to an area of low concentration

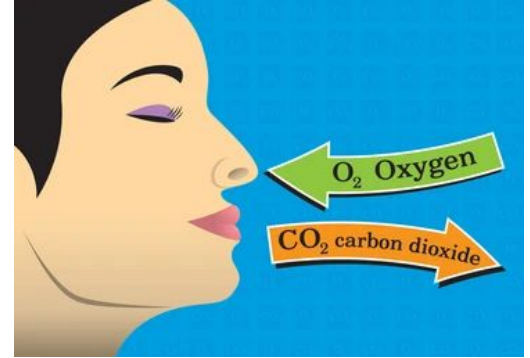


Diffusion of gases

Our body needs oxygen from the air so that cells can carry out the process of **respiration**.

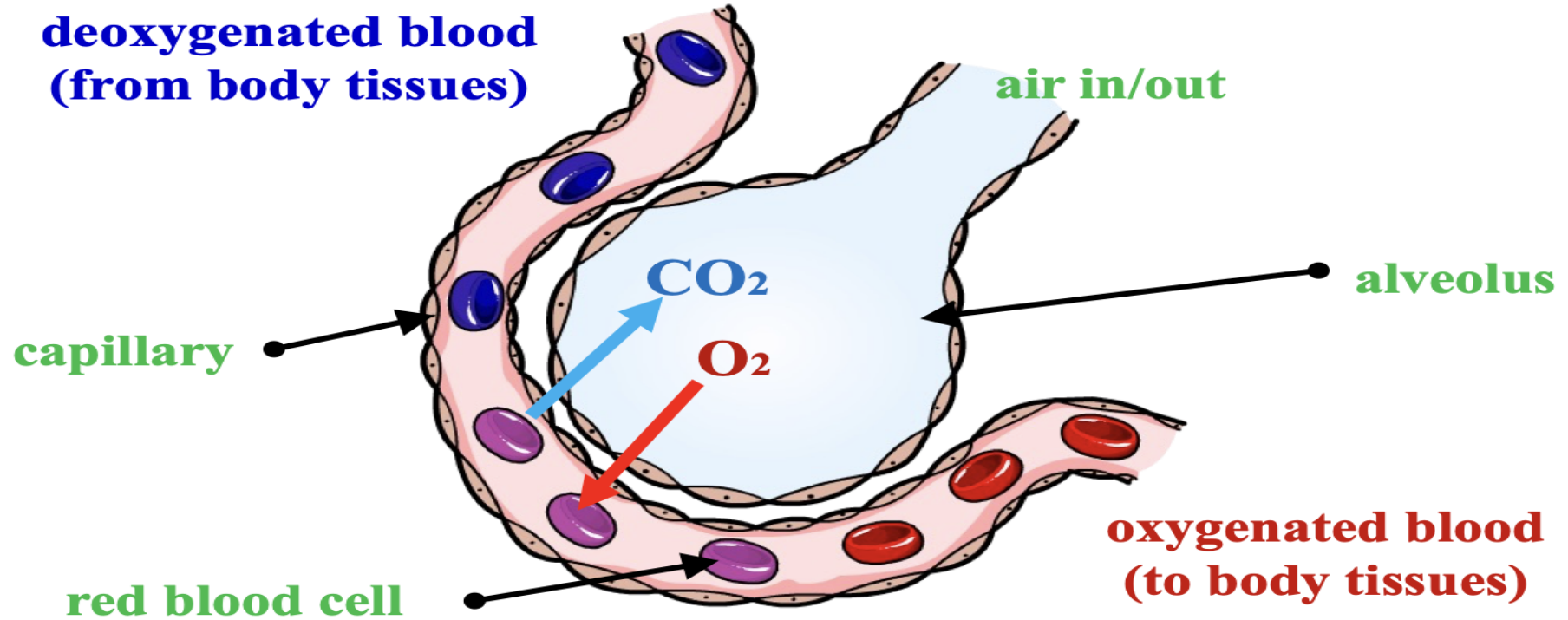
Oxygen **diffuses** from the air into our blood.

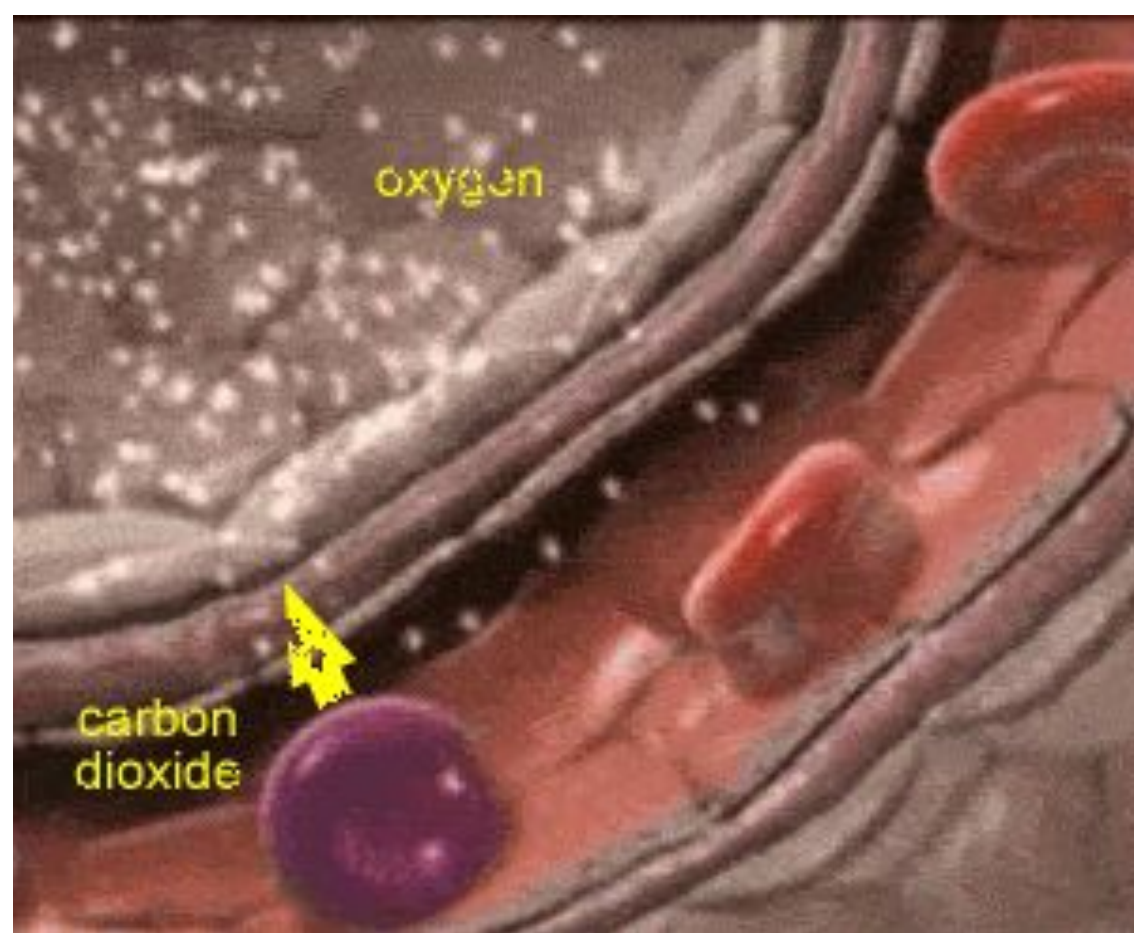
Carbon dioxide **diffuses** out of the blood into the air.



Name the part of the respiratory system that gas exchange occurs

Structure of the alveolus





Describe to your partner what is happening in this animation

Keywords: oxygen, carbon dioxide, alveoli, blood, capillary

Key Points

Copy the sentences into your books and fill in the blanks:

1. Oxygen **diffuses** from the air in the _____ into the _____.
2. Carbon _____ diffuses from the blood into the _____ in the alveoli.
3. We breathe out air. The air we breathe out contains _____ oxygen but _____ carbon dioxide.

Key Points

Copy the sentences into your books and fill in the blanks:

1. Oxygen **diffuses** from the air in the alveoli into the blood.
2. Carbon dioxide diffuses from the blood into the air in the alveoli.
3. We breathe out air. The air we breathe out contains less oxygen but more carbon dioxide.



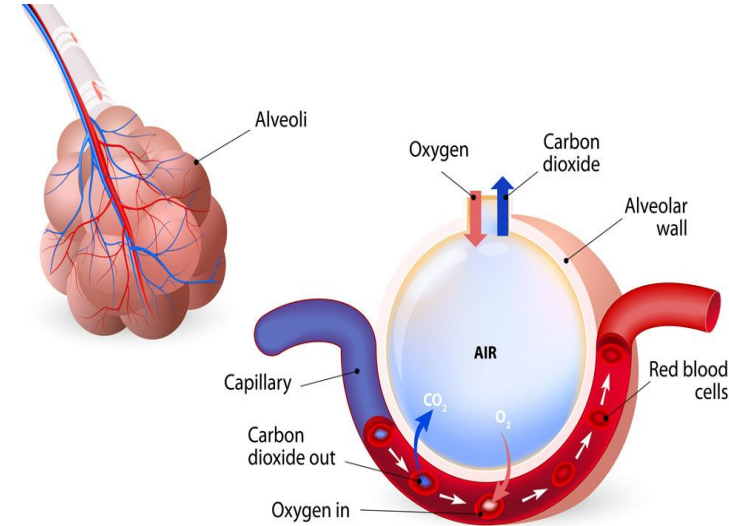
What are the adaptations of the alveoli?

Alveoli Adaptations

Oxygen needs to **diffuses** quickly from the air in the alveoli into the blood.

Alveoli have:

1. Thin walls, made of just one layer of cells
2. A **large surface area**: There are millions of alveoli and they are spherical
3. Many capillaries surrounding them, so they have a **good blood supply**
4. a **moist surface**, which helps gases to diffuse across

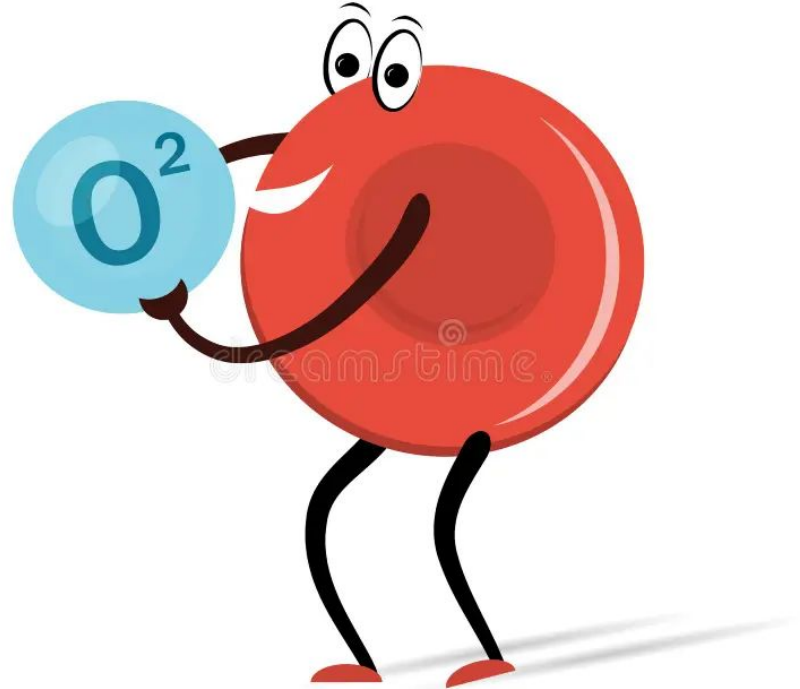


Red blood cells

When oxygen gets into the blood, it **dissolves**.

The oxygen goes into the **red blood cells** where it combines with **haemoglobin**.

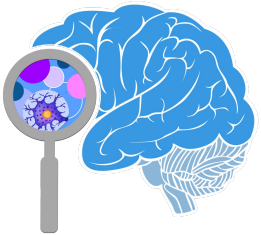
We will learn more about this next week!



Gas exchange continued

Lesson objectives:

- To create a model to demonstrate how surface area affects diffusion.



Why are air sacs so small?

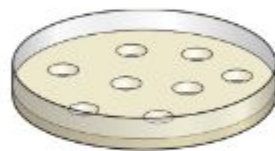
In this activity, you will use some agar jelly to represent the lungs, and some coloured liquid to represent oxygen in the air.

You will need:

- two Petri dishes filled with agar jelly
- two cork-borers, one with a diameter of 10 mm and the other with a diameter of 5 mm
- some coloured dye
- a dropper pipette

Method

- 1 Use the larger cork-borer to make eight holes in the jelly in one of the dishes. Space the holes evenly in the dish.
- 2 Now use the smaller cork-borer to make 32 holes in the jelly in the other dish. Try to space the holes evenly in the dish.
- 3 Using the dropper pipette, carefully fill each hole in both dishes with the coloured dye. Try to put the same quantity of dye into each hole. It's really important not to get any dye on the jelly!
- 4 Leave both dishes for at least 15 minutes.
- 5 Predict what you think will happen.
- 6 After 15 minutes (or a little bit longer if things are happening slowly) record your observations.





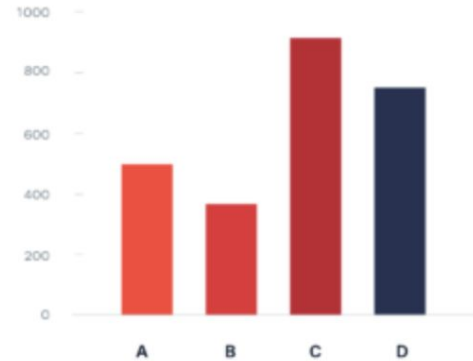
Questions

- 1 The holes that you made in the jelly represent the air sacs in the lungs. The coloured dye represents oxygen in the air sacs. The holes in the jelly are an **analogy** for the air sacs, and the dye is an analogy for oxygen.

Explain how your observations help to show what happens to oxygen in the lungs.

- 2 The total volume of the 32 small holes is the same as the total volume of the eight large holes. Use your observations to suggest why it is better to have a lot of very small air sacs in the lungs, rather than just a few large ones.
- 3 Do you think that the agar jelly with holes is a good model for what happens in the lungs? Explain your answer.

Complete Workbook page 5



Note:

- Make sure you arrange the data accordingly in order (from lowest to highest).
- Draw graph using pencil only!
- Bar chart is the bars not sticking to each other!
- first add up all the numbers in your data set. Then, divide the total sum by the count of the numbers you added.